

Block Chain - Each block that is generated has the hash of the previous block included in it. This creates a link to every coin ever mined in the bitcoin network. When a lot of such blocks come together you get a blockchain. A block chain can also be considered as a history of transactions which is shared by all nodes that are mining or participating in the Bitcoin ecosystem or basically a digital ledger. A complete copy of the bitcoin blockchain will have every movement of every coin ever mined.

Difficulty - About every 2016 blocks that are released, the Bitcoin ecosystem adjusts the difficulty of verifying blocks depending upon the time taken for the previous 2016 blocks. This is roughly considered to be an increment of 20-30% of the previous value. The difficulty parameter is adjusted so as to keep the average bitcoin generation rate static and only one block will be verified on an average of ten minutes for the next 2016 blocks. This “difficulty” parameter is usually expressed as a number [the difficulty at the time of writing this article was 1,789,546,951]. The difficulty is designed to be inversely proportional to the target.

Double-Spending - Trying to spend bitcoins or any digital currency that has already been spent in a previous transaction

Generate Bitcoins - When a miner finally finds a block, it gets virgin bitcoins as a reward. The number of bitcoins awarded for verifying a new block is 25 BTC for each segment of 210,000 blocks. It used to be 50 BTC per block but this number goes down every subsequent 210,000 blocks or so. It has been 4 years since the inception of bitcoins when the first 210,000 blocks were verified and since the rate of generation is constant, it will be another 4 years before the number of BTC awarded is halved again. The total amount of bitcoins that will ever be mined has been calculated to be approximately 21 million BTC.

Hash - The output generated by a hash function is called a hash.

Hash function - It is a computer algorithm which takes any input data and then runs it through a few calculations and shifting processes before generating a fixed length output called the hash. Passwords are commonly stored as a hash so that your actual password is never stored anywhere. Depending on the algorithm used, the probability of two inputs having the same hash does exist. In the Bitcoin system, a hash is difficult to produce but easy to verify and forms the backbone of the whole mining process.

Hash rate - The number of hashes calculated by the hardware is the hash rate. Commonly used indices are Megahashes/second and Gigahashes/second.